Question	Answer	Marks	Partial Marks
9(a)(i)	$\frac{3}{8}$	**2**	**M1** for $8y = 3x + 20$ or better
9(a)(ii)	(0, 2.5) oe	**1**	
(b)(i)	15.6 or 15.62…	**3**	**M2** for $\sqrt{(9 - -3)^2 + (-2 - 8)^2}$ oe seen or **M1** for $(9 - -3)^2$ or $(-2 - 8)^2$ oe seen
9(b)(ii)	$y = -\frac{5}{6}x + 4$ oe	**3**	**M1** for gradient $\frac{-2 - 8}{9 - -3}$ oe **M1** for substituting (6, –1) into a linear equation oe
9(b)(iii)	$y = \frac{6}{5}x - \frac{3}{5}$ oe	**4**	**M1** for gradient -1 / *their* $\left(-\frac{5}{6}\right)$ **B1** for midpoint at (3, 3) **M1** for *their* midpoint substituted into $y = \textit{their } m \times x + c$ oe
10(a)(i)	$x + 5$	**2**	**B1** for linear equation with positive gradient or intercept 5
10(a)(ii)	2 sin x oe	**2**	**B1** for recognition of sin or cos(x – 90)
10(b)	tangent ruled at P	**B1**	
	1.3 to 1.4	**B2**	**dep** on tangent drawn **M1** for rise/run
11(a)	4	**1**	
11(b)	52	**2**	**M1** for $f(8)$ seen or $7 \times \frac{2x}{x-3} - 4$
11(c)	$7x^2 - 4$	**1**	
11(d)	$\frac{7x^2 - 21x + 12}{2(x-3)}$ or $\frac{7x^2 - 21x + 12}{2x - 6}$ final answer	**3**	**M1** for $(7x - 4)(x - 3) + 2 \times 2x$ **B1** for denominator $2(x - 3)$ or $2x - 6$
11(e)	–3	**2**	**M1** for $7x + 14 - 4 = -11$
11(f)	[p =] 0 and [p =] 1	**2**	**B1** for each
12(a)(i)	$\left(-\frac{1}{2}, 4\right)$ and $\left(\frac{1}{2}, 2\right)$	**5**	**B2** for $12x^2 - 3[= 0]$ or **B1** for $12x^2$ or – 3 **M1** for their derivative = 0 or dy/dx = 0 **B1** for [x =] – ½ and ½ or one coordinate pair correct

Question	Answer	Marks	Partial Marks
12(a)(ii)	$\left(-\frac{1}{2}, 4\right)$ Max with reason $\left(\frac{1}{2}, 2\right)$ Min with reason	**3**	**B2** for one correct with reason or **M1** for correct attempt to find e.g. 2nd derivative/gradients/sketch
12(b)	line $y = x + 3$ ruled	**M2**	**B1** for $[y =]x + 3$ identified or rules $y = x + k$ or $y = px + 3$
	–0.7 to –0.8 2.7 to 2.8	**A1**	